

Title

Research on AI involvement in HR Management

Abstract

Artificial Intelligence (AI) is involving in so many industries and automating so many sectors of market. It can be easily seen that Artificial Intelligence has widely used in the development and text generation fields. We can see so many applications that have integrated AI. Some of them are VS code, Copilot, MS office. In this paper we have discussed about the involvement of Artificial Intelligence in the sector of Human Resource Management (HRM). AI integration has already get started for the automation of HR tasks such as JD creation, Emailing, Interviewing candidates, Scheduling etc. AI also has big impact in the analysis sector of HRM. Large amount of data like behaviour, work information can be easily analysed by AI tools to create insights in seconds. This helps the HR to take right decisions at right time.

AI is helping to analyse, automate, generation and take decisions. This has a positive impact on the operations. While there are a large number of benefits of AI involvement there exist some cons of the AI.

This literature highlights the involvement of the AI in human resource management sector and the challenges it faces. This review synthesizes current findings and discusses implication for HR practitioners and researchers.

1. Introduction

1.1 AI IN HRM

Human Resource is evolving with the integration of AI in the sector and transforming traditional procedures [4][7][13]. AI integrated in the foundation of the HR operations [6][11]. AI is having a great capacity of learning, understanding, analysing and creating core insights of the large amount of data that used to take very much time with the manual human actions [11][16][17].

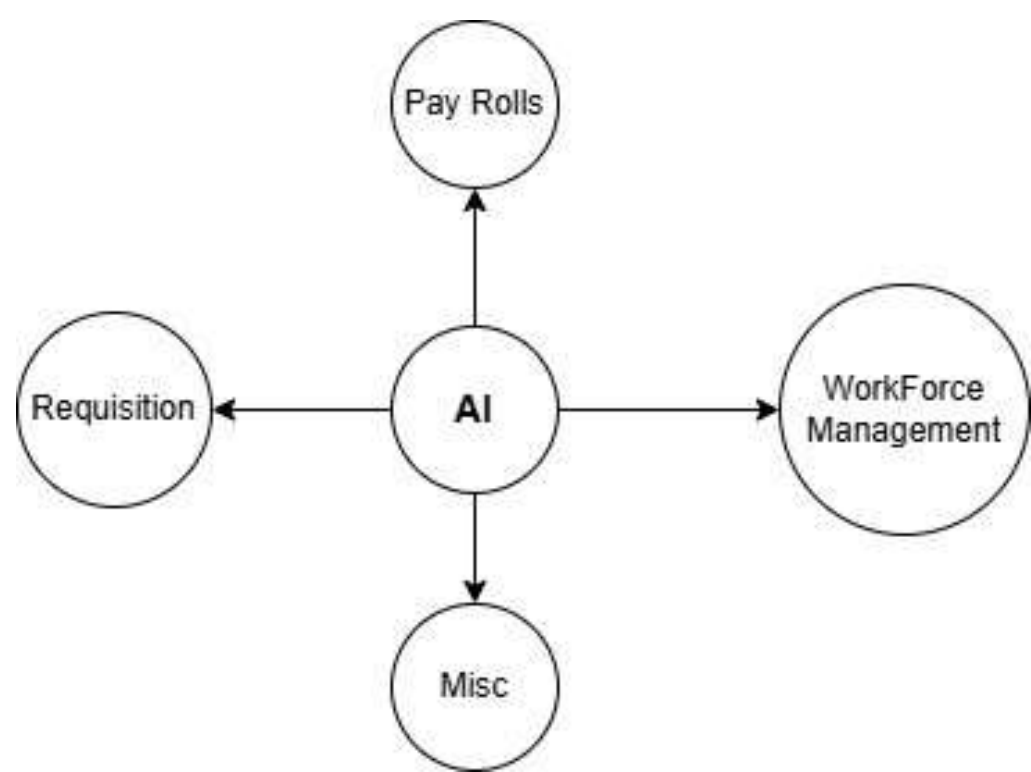


Fig. 1.1

We can see that recent researches exhibit the ‘*exponential growth*’ of AI integration in the Human Resource Management [4][7].

AI is capable of dealing with core operations of the HR [1][9].

These core tasks are:

- Screening Resume [1][12]
- Scheduling Interviews [9][12]
- JD creations [1][9]
- Pay roll management [6][11]
- Interviews [12][18]
- AI generated assessments etc. [5][9]

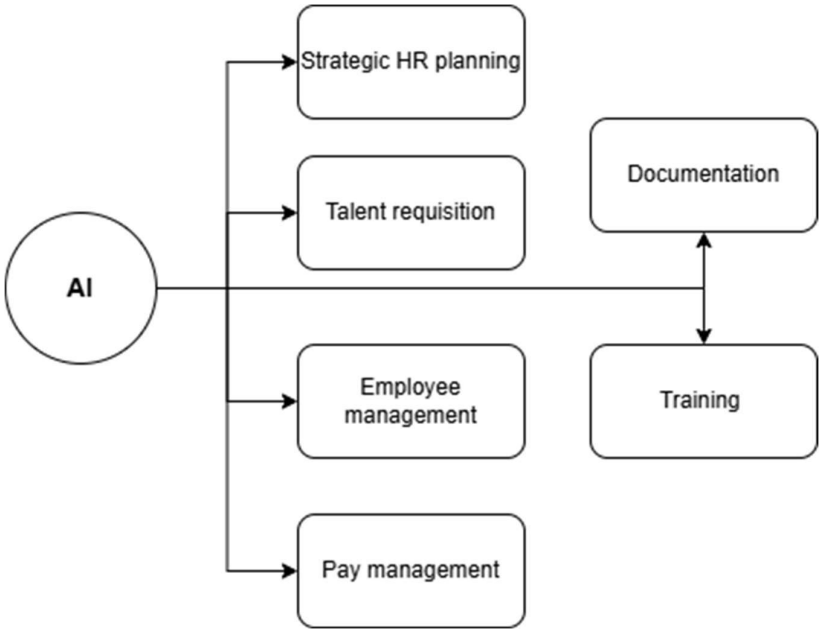


Fig 1.2

As the Artificial Intelligence is evolving and involving in the HR operations researchers have caught the various impacts of the AI. This leads to more accurate, efficient and faster. [4][7][13].

This paper reviews the effects of the AI and what challenges it faces in the particular sector. We will start with the reviews of several research papers and conclude what is going on. [4][7][17].

1.2 MOTIVATION

In today's modern world the AI involvement is increasing rapidly and impacting the sector in its roots. The operations which require in depth manual intervention of human and takes a lot of time of the organization are now being done in such a minor period of time with integration of AI. The AI has evolved enough to bring revolution in the Human Resource Management Sector.

This evolvement of the AI in HRM motivates me to have a research on the AI that is being evolving in the Human Resource Management Sector.

1.3 Background

Use of AI has been increased rapidly in past years. This enables organizations to automate their work using the technology. HR operations like scheduling, monitoring, interviewing, pay roll management get automated with the help of AI.

As there are so many pros of AI we have seen the cons also. AI comes with higher capability of analyzing and as it is trained on data a natural bias also comes with it. Some privacy and security issues are also there.

Analyzing all the features of the AI it creates the foundation for the examining AI by highlighting both its positive and negative sides.

1.3 Contribution

This research makes the following contribution:

- 1.3.1** Comprehensive Mapping of AI with HR processes
- 1.3.2** Comparative analysis of traditional and modern HR practices
- 1.3.3** Identification of challenges and impacts
- 1.3.4** Evaluation of Organizational Impact

2 Methodology

This study adapts the mixed approach for the research. We have been reading the several research papers and surveys as well and concluding based on the information we gather and some personal experiences.

This study uses related research papers and surveys for the gathering of insights then we go to multiple outcomes and filter out main results or findings for the final conclusion.

This based exclusively on academic research papers indexed in reputable databases like:

- Scopus
- Web of Science
- IEEE Xplore
- SpringerLink
- Elsevier

Analytical Approach

The study will used comparative analysis from the papers and will be used to extract final conclusions.

3 Results and Findings

We have found that utilization of Artificial Intelligence increases the throughput and has given better results in the HR functionalities [4][7][11]. Industries having good technical touch are using AI in their daily operations [7][13].

Following table shows the findings for the integration of Artificial Intelligence.in the Human Resource Management Sector.

Criteria	Traditional Recruiting	AI Recruiting
Speed	4-8 weeks	<2 weeks
Quality Hire	Deep interpersonal Insights	Skill + data driven fit
Candidate Experience	Personalized, Slower	Fast, Consistent, but less human
Scalability	Limited by team size	High
Bias & Fairness	Human Bias risk	Reduced bias with monitored algorithms
Cost	High per-hire	Low per-hire

Table 3.1

3. 1 Impacts of AI in HRM

3.3.1 AI and Recruitment

According to multiple surveys we found that AI :

- Reduce time-to-hire significantly [1][12]
- Enhance consistency [1][9]
- Improve candidate screening accuracy [1][12]
- Enhance objectivity in selection [2][15]

Organizations having higher touch with technology have adopted the AI already [12][7]. Importantly, several studies indicates that AI involvement has increased the fair selection records than human selection records [2][15][18].

3.3.2 HR Digitalization and Organizational Agility

Survey findings has shown that AI has a positive impact. It does:

- HR digital transformation [11][4]
- Increased agility of organizations [4][7]
- Increased responsiveness to workforce changes [11][17]

AI-driven tasks like monitoring, real-time feedback, analytics have increased the throughput and efficiency [4][10][11].

3.3.3 Performance management and employee development

AI based performance evaluation system contribute :

- Continuous performance evaluation [5][9]
- Improvised fairness [5][8]
- Transparency in feedback [5][10]
- Support personalization [9][16]

Organizations using AI analytics tool, are saving their lot of resources. [5][16][17].

3.3.4 Workforce analytics and Strategic HR decisions

Researches exhibit that AI improves:

- Talent retention strategies [9][16]
- Long-term workforce planning [16][17]
- Productivity analysis [11][4]

Predictive model show high accuracy in HR decisions [16][11].

3.3.5 Ethical and Human Concerns

Surveys also highlighted challenges related to:

- Algorithmic bias [2][15][18]
- Data privacy [8][6]
- Trust [8][18]

It can be easily seen that the organizations having higher trust among employees who provide transparency and involvement of human [3][8][18].

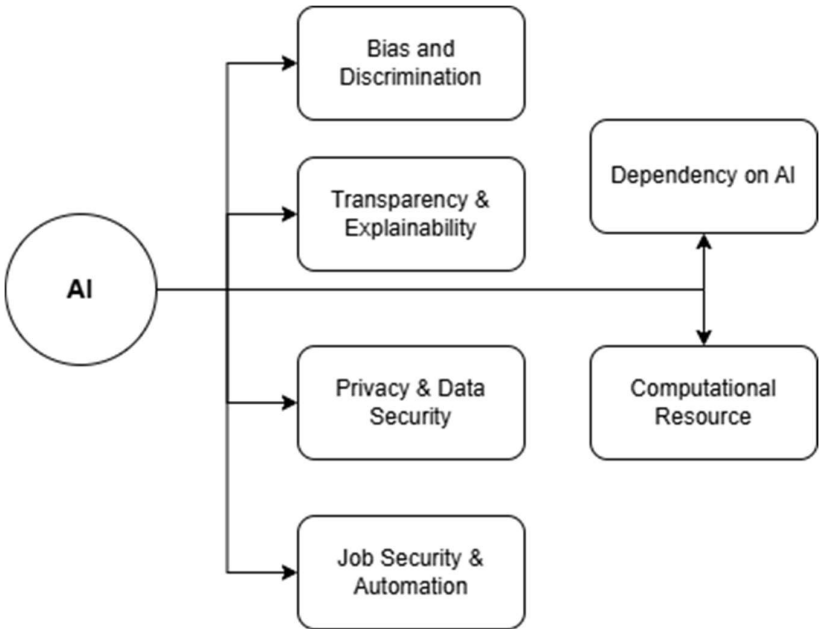


Fig. 3.1

The above diagram describes the key challenges of the AI integration in HR Management sector.

3.2 Efficiency of operations with AI integration

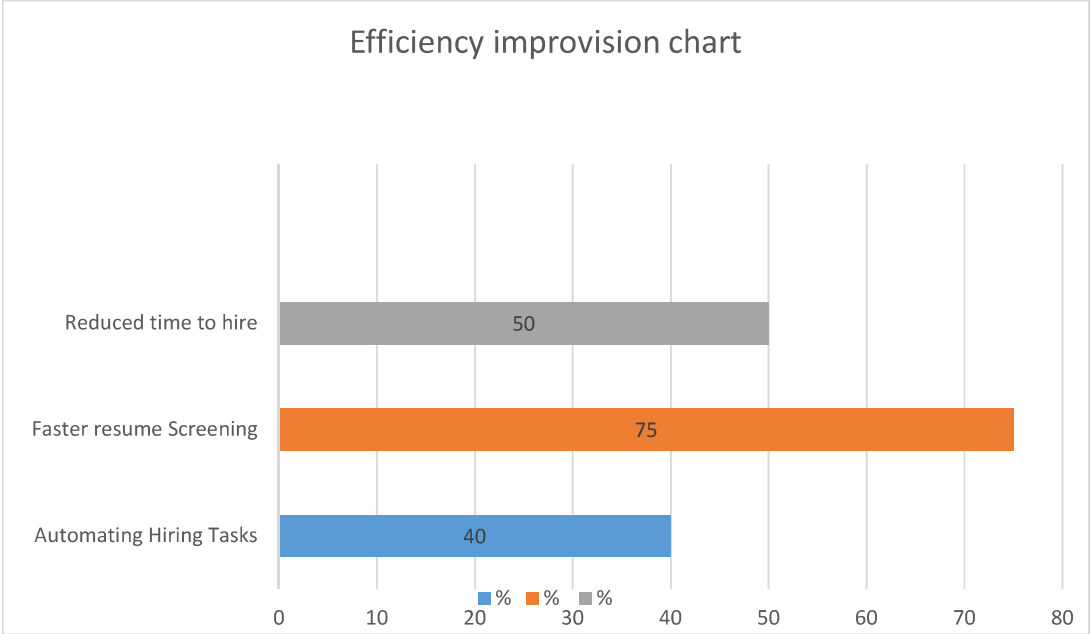


Fig 3.2

Above chart has been obtained from the research findings and is used for comparison basis only [19].

4 Discussion

The findings from the various papers confirm that AI has evolved enough to help humans to improve the various operations. AI integration with human resource management helps in the task of requisition, analytics, notifications, management etc.

The most important finding we get from the analysis of research papers is that **organizations readiness and governance matter more than the technology itself**. AI provide the faster implementation for various tasks but it only succeeded when human involvement is there. Even AI has been integrated to systems but control is still in the hands of humans.

The discussion also highlights a critical paradox:

1. AI reduces human bias in HR decisions, yet
2. Poorly designed algorithms produce biased models which causes biased decisions.

This proves that even AI does all its work with accuracy it is necessary that an expert human should have the accountability of the work and should monitor all the tasks, results which is done by Artificial Intelligence.

Additionally it is seen that even AI integration empower the organization but technically skilled and trained is going to be must for the organizations otherwise organizations will face resistance to adopt AI.

5 Conclusion

This research shows that AI has been evolved and integration of the AI in the HRM increases the throughput and accuracy of the operations. This shows that organizations adopting the AI will have an edge in the market and remaining one will face resistance from the market and will get suffocated with the high cost due to manually powered task.

AI adoption is not that much easy as it seems. To integrate the AI in the existing system is a difficult task that requires complete redesign and restructure. Integration of the AI is not only the main part but upskilling the existing man power is also required.

This paper indicates that the AI in HRM has various pros. This increases the throughput but it also comes with various challenges like biased training of models, unavailability of human emotions, computing resources, trusts etc. We understand that AI is still in evolving state and we need to make sure that integrating it in our systems will be in controlled and secure manner.

6 References

1. Albert, E. T. (2019). Artificial intelligence in talent acquisition: A review of AI applications used in recruitment and selection. *Strategic HR Review*, 18(4), 215–221. <https://doi.org/10.1108/SHR-08-2019-0064>
2. Bogen, M., & Rieke, A. (2018). Help wanted: An examination of hiring algorithms, equity, and bias. *Technology Science*, 1–23.
3. Chamorro-Premuzic, T., Polli, F., & Dattner, B. (2019). Building ethical AI for talent management. *Harvard Business Review*, 97(6), 62–71.
4. Dima, J., Gilbert, M.-H., Dextras-Gauthier, J., & Giraud, L. (2024). The effects of artificial intelligence on human resource activities and the roles of the human resource triad. *Frontiers in Psychology*, 15, 1360401. <https://doi.org/10.3389/fpsyg.2024.1360401>
5. Fernández-Martínez, E., García-Pérez, A., & Valencia-García, R. (2022). Artificial intelligence techniques for employee performance evaluation: A systematic review. *Applied Sciences*, 12(6), 3052. <https://doi.org/10.3390/app12063052>
6. Jatobá, M. A., Bellas, F., de Oliveira, J. P., et al. (2019). Artificial intelligence in human resources management: Challenges and applications. *IEEE Technology and Society Magazine*, 38(4), 25–34. <https://doi.org/10.1109/MTS.2019.2941188>
7. Kshetri, N. (2021). Artificial intelligence in human resource management: Opportunities and challenges. *IEEE IT Professional*, 23(3), 5–9. <https://doi.org/10.1109/MITP.2021.3058461>
8. Leicht-Deobald, U., Busch, T., Schank, C., et al. (2019). The challenges of algorithm-based HR decision-making. *Journal of Business Ethics*, 160(2), 377–392. <https://doi.org/10.1007/s10551-019-04225-w>
9. Madanchian, M. (2024). From recruitment to retention: AI tools for human resource decision-making. *Applied Sciences*, 14(24), 11750. <https://doi.org/10.3390/app142411750>
10. Meijerink, J., Bondarouk, T., & Lepak, D. (2021). Virtual HRM and employee engagement: The role of digital HR practices. *Human Resource Management Review*, 31(2), 100783. <https://doi.org/10.1016/j.hrmr.2020.100783>
11. Murugesan, U., Subramanian, P., Srivastava, S., & Dwivedi, A. (2023). Artificial intelligence impacts on human resource digitalization in Industry 4.0. *Decision Analytics Journal*, 7, 100249. <https://doi.org/10.1016/j.dajour.2023.100249>
12. Pan, Y., Froese, F. J., Liu, N., Hu, Y., & Ye, M. (2021). The adoption of artificial intelligence in employee recruitment: The influence of contextual factors. *International Journal of Human Resource Management*, 33(7), 1–23. <https://doi.org/10.1080/09585192.2021.1879206>
13. Parry, E., & Battista, V. (2019). The impact of emerging technologies on work: A review of the evidence and implications for HR. *Human Resource Management Journal*, 29(4), 1–16. <https://doi.org/10.1111/1748-8583.12226>
14. Pillai, R., Sivathanu, B., & Dwivedi, Y. K. (2020). Shopping intention at AI-powered automated retail stores. *Journal of Retailing and Consumer Services*, 57, 102207. <https://doi.org/10.1016/j.jretconser.2020.102207>
15. Raghavan, M., Barocas, S., Kleinberg, J., & Levy, K. (2020). Mitigating bias in algorithmic hiring: Evaluating claims and practices. *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency*, 469–481. <https://doi.org/10.1145/3351095.3372828>

16. Tursunbayeva, A., Di Lauro, S., & Pagliari, C. (2018). People analytics—A scoping review of conceptual boundaries and value propositions. *International Journal of Information Management*, 43, 224–247. <https://doi.org/10.1016/j.ijinfomgt.2018.08.002>
17. Úbeda-García, M., Marco-Lajara, B., Zaragoza-Saez, P. C., & Poveda-Pareja, E. (2025). Artificial intelligence, knowledge, and human resource management: Strategic implications. *Journal of Innovation & Knowledge*, 10, 100809. <https://doi.org/10.1016/j.jik.2025.100809>
18. van den Broek, E., Sergeeva, A., & Huysman, M. (2021). Hiring algorithms: An ethnography of fairness in practice. *Organization Studies*, 42(5), 789–815. <https://doi.org/10.1177/0170840620928660>
19. <https://www.yomly.com/ai-in-hr-statistics>